Self-supervised models of human texture-discrimination

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How do we discriminate same/diff stuff?

- We interpret scenes by segmenting them into same/different stuff.
- Texture is an important bottom-up cue that we use to do this.
- How do we do this so quickly on new scenes with novel textures, with a foveated vision?

Method: human experiment and two models

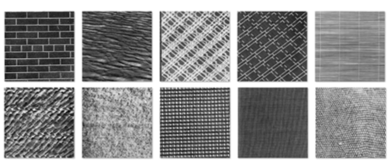
- We measure human responses to say if two arbitrary texture patches at the fovea or periphery are the same or different.
- We build a Bayesian model that does this task based on three texture properties: power spectrum, colour histogram, and texture edge properties.
- We also build a CNN to do the same task.
- We run these models at different eccentricities, by blurring and downsampling the patches.
- We explore self-supervision: learning texture properties by bootstrapping simpler visual cues like proximity and colour from natural images.

Conclusions

- Our Bayesian model can use a boundary on 3 texture features to discriminate general arbitrary textures
- It also performs very similar to humans across eccentricities, and the pattern of human errors aligns with it.
- Our self-supervised Bayesian model learns a similar boundary as when trained with explicit textures.
- Our self-supervised CNN performs with 76% accuracy on textures.

Experiment

- We choose the Brodatz texture set:



- We pick small 1°×1° patches from these textures. They are small since we want to study the local bottom-up process in the early visual system, where receptive fields are of this size.
- We show two patches at the fovea, or at 1.6°/5°/11.5°, where our resolution drops by 2x/4x/8x from the fovea:

- Subject can see them for 200ms, then has to say if they are the same or different texture.

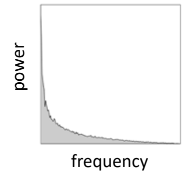
We build a Bayesian model using three texture properties

We build a Bayesian model for this texture discrimination task.

Given the two texture patches, we compute three Bayesian decision variables based on the statistics of three texture properties.

1. Power spectrum

- The spatial power spectrum $p(f)$ of natural images and textures is similar to that of filtered noise, which has an **independent exponential distribution** at each frequency f.
- Using this we can make a Bayesian decision variable (log likelihood ratio) that the two patches have same or different power spectrum:

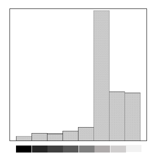


$$L_P = \sum_f \ln \frac{[\hat{p}_a(f) + \hat{p}_b(f)]^2}{4\hat{p}_a(f)\hat{p}_b(f)}$$

- The information captured by the power spectrum is similar to that contained in the population response of complex cells covering the patch region.

2. Grey/colour histogram

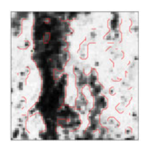
- An important texture feature is the distribution of greyscale or colour values across them.
- Assume that a texture has fixed probabilities of these values. The pixel values in a patch are sampled from them. This produces a **multinomial distribution** of the histogram counts N_i of pixel values.
- Using this we can make a Bayesian decision variable (log likelihood ratio) that the two patch histograms are from the same or different distribution :



$$L_H = \sum_i N_{ai} \ln \frac{N_{ai}}{N_a} + N_{bi} \ln \frac{N_{bi}}{N_b} - \sum_i (N_{ai} + N_{bi}) \ln \frac{N_{ai} + N_{bi}}{N_a + N_b}$$

3. Edge properties

- We detect edge pixels on the texture patch, by computing and thresholding the gradient.
- Then we measure different properties of these edge elements, and compute likelihood ratios that they came from same or different distributions:



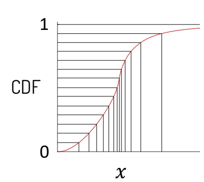
- The number of edge pixels (N_a or N_b) in the patch (of N pixels) are modeled as a **binomial** with an underlying probability. This gives us a log likelihood ratio:

$$L_N = N_a \ln\left(\frac{2N_a}{N_a + N_b}\right) + N_b \ln\left(\frac{2N_b}{N_a + N_b}\right)$$
$$+(N - N_a) \ln \frac{2(N - N_a)}{2N - N_a - N_b} + (N - N_b) \ln \frac{2(N - N_b)}{2N - N_a - N_b}$$

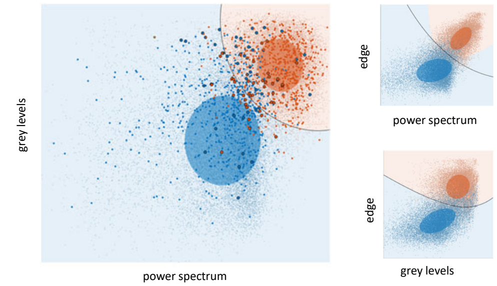
- Given the total number of edge pixels, we now model the histograms of their sharpness (gradient magnitude) and orientation as multinomial. This gives us two log likelihood ratios L_M and L_O.
- The overall decision variable based on the edges is $L_E = L_N + L_M + L_O$.

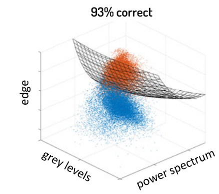
Efficient coding improves our model

- **PCA:** For colour textures, we perform PCA on the RGB values across natural images, and map them to more independent channels: luminance/blue-yellow/red-green.
- **Histogram equalization:** the histogram bins of grey/colour levels and edge properties are adjusted so that there are equal counts in each bin.



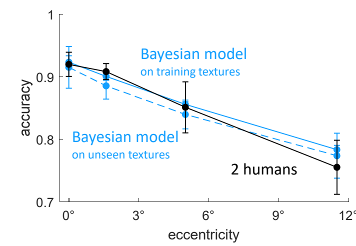
Bayesian model can discriminate general textures





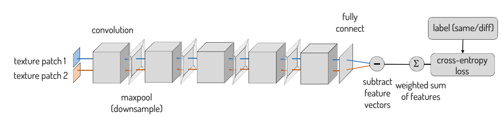
- The Bayesian model can use a single boundary to discriminate general arbitrary textures with 93% accuracy.
- The errors made by humans in the experiments are larger dots outlined in black.
- They are closer to the boundary, so our Bayesian model is also somewhat aligned with humans.

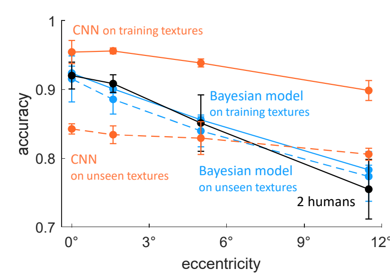
Bayesian model matches human performance



The model performs equally well on unseen textures, so it can discriminate general arbitrary textures

We also build a CNN to discriminate textures





- Currently it may be over-learning the training textures, and generalizing poorly.
- So we will train it on more texture sets: fabric, ALOT, DTD and T1K+.

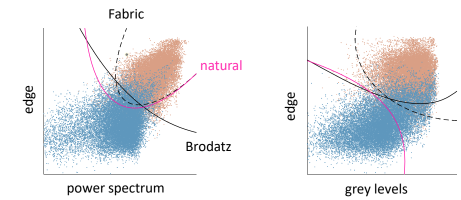
Self-supervised learning

Broad idea

- Animals do not get vast amounts of labelled same/different textures to learn discrimination
- How then do our visual systems naturally know or learn this?
- Nearby patches in a scene of the same colour are likely to be the same stuff.
- A simple visual system can use this as a rough starting point to label same/diff stuff.
- Using these self-generated labels, it can then learn texture discrimination
- It can then use texture too to discriminate same/different stuff
- Even if many of the simple-cue-based labels are wrong, the errors are random, so it will still learn the correct texture features

Our method

- We test this theory with our **natural images dataset**.
- We pick nearby patches with similar **perceptually-uniform LAB colour** as 'same' pairs
- We pick patches that are far apart with different LAB colours as 'different' pairs
- Then we remove the colour, and use our Bayesian model to compute their texture-features and learn a boundary
- This produces a very similar boundary, as when explicitly trained on Brodatz and fabric textures.



- So evolutionary or lifetime learning can use vast amounts of self-generated data to learn correct discrimination features.
- Our CNN trained on these self-labelled natural image patches achieves 76% accuracy on the Brodatz textures.

Future plans

- Conduct discrimination experiments where one patch is at the fovea and the other is at a different eccentricity.
- Model the above task by 'moving' the fovea patch to the periphery by blurring and downsampling, then comparing the two.

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